Sendov's Conjecture in Degree Nine

Principia Math

Abstract

We prove Sendov's conjecture for polynomials of degree nine. After fixing a zero $a \in [0, 1]$, a hypothetical counterexample gives two reciprocal configurations, one formed from the remaining zeros and one from the critical points. Their common first moment yields a strong localization inequality. A symmetric integral identity then reduces the remaining argument to positivity of explicit polynomials. The range $a \le 9/20$ is elementary; the interval $9/20 \le a \le 9/10$ is covered by eighteen exact Bernstein certificates; and the boundary range $9/10 \le a \le 1$ is handled by a rescaled single certificate. All finite computations are performed in exact rational arithmetic.

1 Introduction

Sendov's conjecture asserts that if all zeros of a complex polynomial lie in the closed unit disk, then the closed unit disk centered at each zero contains a zero of the derivative. We prove the degree-nine case.

Theorem 1.1. *Let*

$$p(z) = \prod_{k=1}^{9} (z - z_k), \qquad |z_k| \le 1.$$

For every zero a of p, there is a zero ζ of p' such that

$$|a - \zeta| \le 1.$$

The proof is computer-assisted only at two finite positivity checks. Both are certified by expanding explicit rational polynomials in tensor-product Bernstein bases and checking every coefficient by exact arithmetic. The analytic reduction and the certificate polynomials are given below; the ancillary verifier contains no floating-point operations.

By rotation we may take the distinguished zero to be a real number $a \in [0, 1]$. We argue by contradiction, assuming throughout that every zero of p' has distance strictly greater than 1 from a.

2 Two preliminary lemmas

We use the Grace–Walsh–Szegő coincidence theorem in its standard form: if a symmetric multi-affine polynomial is evaluated at points in a convex circular region, then the same value is attained on the diagonal at a point of that region; see, for example, [1, Proposition 1].

Lemma 2.1 (separation from a zero). *Let P be a polynomial of degree $n \ge 2$, let α be a zero of P, and suppose that*

$$|\alpha - \zeta| > \rho$$

for every zero ζ of P'. Then α is simple and every other zero β of P satisfies

$$|\beta - \alpha| > 2\rho \sin \frac{\pi}{n}.$$

Proof. The simplicity of α is immediate. Write the zeros of P' as $\zeta_1, \dots, \zeta_{n-1}$ and set $w = \beta - \alpha$. Since $P(\beta) = P(\alpha)$,

$$0 = \int_0^1 \frac{P'(\alpha + tw)}{P'(\alpha)}\, dt = \int_0^1 \prod_{j=1}^{n-1} (1 - tu_j)\, dt, \qquad u_j = -\frac{w}{\alpha - \zeta_j}.$$

The symmetric multi-affine polynomial

$$F(u_1, \dots, u_{n-1}) = \int_0^1 \prod_{j=1}^{n-1} (1 - tu_j)\, dt$$

has diagonal

$$F(u, \dots, u) = \frac{1 - (1-u)^n}{nu},$$

with the value 1 at $u = 0$. Its diagonal zeros are $1 - \mathrm{e}^{2\pi ik/n}$, $1 \le k < n$, and hence have modulus at least $2\sin(\pi/n)$. The coincidence theorem therefore implies that F is nonzero whenever all $|u_j| < 2\sin(\pi/n)$. If $|w| \le 2\rho \sin(\pi/n)$, then the strict critical-point separation gives $|u_j| < 2\sin(\pi/n)$ for every j, a contradiction. $\square$

The second lemma is a one-dimensional convexity estimate.

Lemma 2.2. *Let $0 < m < M$, let $r_1, \dots, r_N \in [m, M]$, and suppose*

$$\prod_{k=1}^{N} r_k \ge C > m^N.$$

Let ν be the least integer such that $M^\nu m^{N-\nu} \ge C$. Then

$$\sum_{k=1}^{N} \frac{1}{r_k^2} \le \frac{N-\nu}{m^2} + \frac{\nu - 1}{M^2} + \left(\frac{M^{\nu-1} m^{N-\nu}}{C}\right)^2.$$

Proof. Put $r_k = \mathrm{e}^{t_k}$. Since $t \mapsto \mathrm{e}^{-2t}$ is decreasing, we may first reduce $\sum t_k$ to $\log C$. On the resulting section of the box $[\log m, \log M]^N$, the convex function $\sum \mathrm{e}^{-2t_k}$ is maximized at an extreme point. Thus all but at most one coordinate are endpoints. By the definition of ν, the maximizing configuration has $N - \nu$ entries equal to m, $\nu - 1$ entries equal to M, and one entry equal to $C/(M^{\nu-1} m^{N-\nu})$. Substitution gives the result. $\square$

3 The reciprocal reduction

Write

$$p(z) = (z - a) \prod_{k=1}^{8} (z - z_k), \qquad p'(z) = 9 \prod_{j=1}^{8} (z - \zeta_j),$$

where multiplicities are retained. A counterexample has a simple. Set

$$X_k = \frac{1}{a - z_k}, \qquad Y_j = \frac{1}{a - \zeta_j}, \qquad \mu = \frac{1}{8} \sum_{j=1}^{8} Y_j = u + iv.$$

Evaluation of p' and p''/p' at a gives the two identities

$$9\prod_{j=1}^{8}(a-\zeta_j)=\prod_{k=1}^{8}(a-z_k), \qquad \sum_{j=1}^{8}Y_j=2\sum_{k=1}^{8}X_k. \tag{1}$$

In particular,

$$\mu=\frac{1}{4}\sum_{k=1}^{8}X_k, \qquad |\mu|<1. \tag{2}$$

Put

$$r_k=|a-z_k|, \qquad \sigma=\sum_{k=1}^{8}r_k^{-2}.$$

By Lemma 2.1 and the unit-disk hypothesis,

$$2\sin\frac{\pi}{9}<r_k\le 1+a. \tag{3}$$

Moreover, (1) and the assumed critical-point separation imply

$$\prod_{k=1}^{8}r_k>9. \tag{4}$$

For exact arithmetic we use

$$m_0:=\frac{681}{1000}<2\sin\frac{\pi}{9}. \tag{5}$$

Indeed, $T=2\sin(\pi/9)$ is the root in $(0,1)$ of $3T-T^3=\sqrt{3}$, the function $3t-t^3$ is increasing on $[0,1]$, and

$$3-\left(3m_0-m_0^3\right)^2=\frac{168535344459219919}{10^{18}}>0.$$

The unit-disk condition also controls the real part of μ. Since $z_k=a-X_k^{-1}$,

$$2a\Re X_k+(1-a^2)|X_k|^2=1+(1-|z_k|^2)|X_k|^2\ge 1.$$

Summing and using (2), we obtain the basic localization inequality

$$8au+(1-a^2)\sigma\ge 8. \tag{6}$$

We shall also need a uniform expansion about the mean μ. Put

$$\eta^2=1-|\mu|^2, \qquad D_j=Y_j-\mu.$$

Then $\sum_j D_j=0$ and

$$\sum_{j=1}^{8}|D_j|^2=\sum_{j=1}^{8}|Y_j|^2-8|\mu|^2\le 8\eta^2.$$

If $e_m(D)$ denotes the mth elementary symmetric function of the D_j, then

$$|e_m(D)|\le c_m\eta^m \qquad (2\le m\le 8), \tag{7}$$

where

$$(c_2, \ldots, c_8) = \left(4, \frac{264}{35}, 24, 56, 28, 8, 1\right), \qquad (\ell_2, \ldots, \ell_8) = (3, 2, 2, 1, 1, 0, 0). \tag{8}$$

For completeness, Newton's identities give $|e_2| \le 4\eta^2$, $|e_3| \le 16\sqrt{2}\eta^3/3 < 264\eta^3/35$, and $|e_4| \le 24\eta^4$. For $m \ge 5$, Maclaurin's inequality applied to $|D_1|, \ldots, |D_8|$, followed by Cauchy–Schwarz, gives $|e_m(D)| \le \binom{8}{m}\eta^m$.

Finally, define

$$G(t) = \prod_{j=1}^{8}(1 - tY_j) = \frac{p'(a-t)}{p'(a)}, \qquad I = \int_0^a G(t)\,dt. \tag{9}$$

Then

$$I = -\frac{p(0)}{p'(a)}, \qquad |I| < \frac{a}{9}, \tag{10}$$

because $|p(0)| \le a$ and $|p'(a)| > 9$. The remainder of the proof derives the opposite inequality.

4 The range $0 \le a \le 9/20$

The case $a = 0$ is already impossible: (4) contradicts $r_k = |z_k| \le 1$. Suppose $0 < a \le 9/20$. Apply Lemma 2.2 with $N = 8$, $C = 9$, $m = m_0$, and $M = 29/20$. Here $\nu = 7$, and exact arithmetic gives

$$\sigma \le \frac{1}{m_0^2} + \frac{6}{(29/20)^2} + \left(\frac{(29/20)^6 m_0}{9}\right)^2 < \frac{1101}{200} =: S_0. \tag{11}$$

By (6),

$$u > \frac{8 - S_0(1-a^2)}{8a}.$$

The right-hand side exceeds 1 precisely when

$$S_0 a^2 - 8a + 8 - S_0 > 0.$$

This quadratic is decreasing on $[0, 9/20]$, and at $a = 9/20$ it equals $781/80000$. Thus $u > 1$, contradicting $|\mu| < 1$.

5 The range $9/20 \le a \le 9/10$

We first isolate the analytic statement certified on each subinterval. For $S > 4$ set

$$L_S(a) = \frac{8 - S(1-a^2)}{8a},$$

and, for $0 \le v \le 1$, define

$$q_S(a, \eta; v) = 1 - \frac{8 - S + Sa^2}{4}v + a^2(1-\eta^2)v^2, \tag{12}$$

$$h_S(a, \eta) = \left(\frac{S}{4} - 1\right)(1 - a^2) - a^2\eta^2. \tag{13}$$

Using the constants in (8), put

$$E_S(a,\eta) = \sum_{m=2}^{8} c_m a^{m+1} \eta^m \int_0^1 v^m q_S(a,\eta;v)^{\ell_m}\, dv \tag{14}$$

and

$$P_{S,\lambda}(a,\eta) = \frac{1-a}{9} + \frac{\eta^2}{18} - \frac{h_S(a,\eta)^4}{9\lambda} - E_S(a,\eta). \tag{15}$$

Since the exponents ℓ_m are nonnegative integers, $P_{S,\lambda}$ is a polynomial with rational coefficients whenever S and λ are rational.

Proposition 5.1 (interval certificate)**.** *Let $0 < \alpha \le \beta < 1$, and let S, λ, Y be positive numbers such that, for every $a \in [\alpha, \beta]$,*

$$\sigma \le S, \qquad L_S(a) \ge \lambda, \qquad Y^2 \ge 1 - \lambda^2, \qquad 2\lambda \ge \beta, \qquad 0 \le \left(\frac{S}{4} - 1\right)(1-\alpha^2) \le 1. \tag{16}$$

If

$$P_{S,\lambda}(a,\eta) > 0 \qquad (\alpha \le a \le \beta,\ 0 \le \eta \le Y),$$

then no counterexample exists with $a \in [\alpha, \beta]$.

Proof. By (6), $u \ge L_S(a) \ge \lambda$, and hence $0 \le \eta^2 = 1 - |\mu|^2 \le 1 - \lambda^2 \le Y^2$. Let

$$J = \int_0^a (1 - t\mu)^8\, dt.$$

For $0 \le t \le a$, we have

$$|1 - t\mu|^2 = 1 - 2tu + (1-\eta^2)t^2 \le 1 - 2tL_S(a) + (1-\eta^2)t^2.$$

After $t = av$, the expression on the right is $q_S(a,\eta;v)$. It is nonnegative, and

$$q_S(a,\eta;v) - 1 \le av(-2\lambda + \beta) \le 0.$$

Thus $0 \le q_S \le 1$. Expanding

$$G(t) = (1-t\mu)^8 + \sum_{m=2}^{8} (-t)^m e_m(D)(1-t\mu)^{8-m}$$

and using (7), with half-integral powers rounded down by $q_S^{5/2} \le q_S^2$, $q_S^{3/2} \le q_S$, and $q_S^{1/2} \le 1$, gives

$$|I - J| \le E_S(a,\eta). \tag{17}$$

Since

$$J = \frac{1 - (1-a\mu)^9}{9\mu},$$

write $R = |1 - a\mu|$. The bound $u \ge L_S(a)$ gives

$$R^2 \le h_S(a,\eta).$$

For a feasible pair (a, η), this implies $h_S \geq 0$, while (16) gives $h_S \leq 1$. Therefore

$$|J| \geq \frac{1 - R^9}{9|\mu|} \geq \frac{1}{9\sqrt{1-\eta^2}} - \frac{h_S(a,\eta)^4}{9\lambda} \geq \frac{1}{9} + \frac{\eta^2}{18} - \frac{h_S(a,\eta)^4}{9\lambda}.$$

Together with (17), this yields

$$|I| - \frac{a}{9} \geq P_{S,\lambda}(a,\eta) > 0,$$

contrary to (10). □

We now give the exact finite certificate. In each row of Table 1, Lemma 2.2, with $M = 1 + \beta$, gives $\sigma \leq S$. The inequality $L_S(a) \geq \lambda$ is equivalent to

$$Sa^2 - 8\lambda a + 8 - S \geq 0,$$

which is checked exactly on the indicated interval. The remaining elementary conditions in (16) are direct rational inequalities.

Table 1: Exact constants for the middle range.

$[\alpha, \beta]$	S	λ	Y
$[9/20, 19/40]$	11043/2000	49/50	199/1000
$[19/40, 1/2]$	2783/500	239/250	587/2000
$[1/2, 21/40]$	2257/400	1863/2000	91/250
$[21/40, 11/20]$	1151/200	1811/2000	849/2000
$[11/20, 23/40]$	11819/2000	879/1000	477/1000
$[23/40, 3/5]$	764/125	1699/2000	66/125
$[3/5, 5/8]$	637/100	817/1000	577/1000
$[5/8, 13/20]$	13093/2000	401/500	239/400
$[13/20, 27/40]$	13113/2000	81/100	1173/2000
$[27/40, 7/10]$	3289/500	409/500	1151/2000
$[7/10, 29/40]$	529/80	413/500	141/250
$[29/40, 3/4]$	6661/1000	1669/2000	1103/2000
$[3/4, 31/40]$	1681/250	843/1000	269/500
$[31/40, 4/5]$	1701/250	213/250	131/250
$[4/5, 33/40]$	2761/400	1723/2000	127/250
$[33/40, 17/20]$	14041/2000	109/125	49/100
$[17/20, 7/8]$	7161/1000	221/250	187/400
$[7/8, 9/10]$	3663/500	359/400	883/2000

Lemma 5.2 (middle-range certificate)**.** *For every row $(\alpha, \beta, S, \lambda, Y)$ of Table 1,*

$$P_{S,\lambda}(a,\eta) > \frac{1}{2100} \qquad (\alpha \leq a \leq \beta,\ 0 \leq \eta \leq Y).$$

Proof. Substitute

$$a = \alpha + (\beta - \alpha)U, \qquad \eta = YV, \qquad 0 \leq U, V \leq 1.$$

The resulting polynomial has bidegree at most $(9, 8)$. Its exact Bernstein expansion is

$$\sum_{i=0}^{9}\sum_{j=0}^{8} b_{ij}\binom{9}{i}U^i(1-U)^{9-i}\binom{8}{j}V^j(1-V)^{8-j}.$$

Across all eighteen rows and all 1620 coefficients, the least coefficient is

$$\frac{1525995182936088798318336809}{3199338912963867187500000000000} > \frac{1}{2100}.$$

The Bernstein basis functions are nonnegative and sum to 1, proving the claim. The expansion and all preceding interval inequalities are verified in exact rational arithmetic by the ancillary file. □

Proposition 5.1 and Lemma 5.2 exclude the entire interval $9/20 \le a \le 9/10$.

6 The boundary range $9/10 \le a \le 1$

At $a = 1$, (6) gives $u \ge 1$, contradicting $|\mu| < 1$. Suppose henceforth that $9/10 \le a < 1$, and put

$$\delta = 1 - a.$$

Applying Lemma 2.2 with $M = 2$ gives

$$\sigma \le \frac{3}{m_0^2} + 1 + \frac{256m_0^6}{81} < \frac{39}{5}. \tag{18}$$

It follows from (6) that

$$1 - u \le \delta\frac{39a-1}{40a} \le \frac{19}{20}\delta. \tag{19}$$

Consequently

$$0 < \eta^2 = 1 - |\mu|^2 \le \frac{19}{10}\delta - \frac{361}{400}\delta^2. \tag{20}$$

Write

$$\delta = x^2, \qquad \eta = xy.$$

Then

$$0 < x < \frac{8}{25}, \qquad 0 \le y < \frac{7}{5}. \tag{21}$$

For $0 \le t \le 1 - x^2$, set

$$q(x, y; t) = (1-t)^2 + x^2t\left(\frac{19}{10} - y^2t\right). \tag{22}$$

By (19), $|1 - t\mu|^2 \le q(x, y; t)$, and the same elementary calculation as before gives $0 \le q \le 1$ on the feasible set. Define

$$\mathcal{E}(x, y) = \sum_{m=2}^{8} c_m x^{m-2}y^m \int_0^{1-x^2} t^m q(x, y; t)^{\ell_m}\, dt. \tag{23}$$

The centered expansion and (7) yield

$$|I - J| \le \delta\mathcal{E}(x, y), \qquad J = \int_0^a (1-t\mu)^8\, dt. \tag{24}$$

Let $R = |1 - a\mu|$. From (19),

$$R^2 \le \delta^2 + \frac{19}{10}(1-\delta)\delta - (1-\delta)^2\eta^2 \le \frac{19}{10}\delta.$$

Also $|\mu| \ge 181/200$. Hence

$$|J| \ge \frac{1}{9} + \frac{\eta^2}{18} - \frac{R^9}{9|\mu|} \ge \frac{1}{9} + \frac{\eta^2}{18} - \frac{\delta}{1000}.$$

For the last inequality, it is enough, by monotonicity in $0 < \delta \le 1/10$, to check

$$\frac{200}{1629}\left(\frac{19}{10}\right)^{9/2}\left(\frac{1}{10}\right)^{7/2} \le \frac{1}{1000};$$

after squaring this is the exact integer inequality

$$200^2 19^9 < 1629^2 10^{10}.$$

Combining this with (24) gives

$$|I| - \frac{a}{9} \ge \delta\,\mathcal{P}(x,y), \tag{25}$$

where

$$\mathcal{P}(x,y) = \frac{1}{9} + \frac{y^2}{18} - \frac{1}{1000} - \mathcal{E}(x,y). \tag{26}$$

This is a rational polynomial of bidegree $(24, 8)$.

Lemma 6.1 (boundary certificate)**.** *For*

$$0 \le x \le \frac{8}{25}, \qquad 0 \le y \le \frac{7}{5},$$

one has

$$\mathcal{P}(x,y) > \frac{1}{100}.$$

Proof. Put $x = (8/25)U$ and $y = (7/5)V$. In the tensor-product Bernstein basis of bidegree $(24, 8)$, all 225 coefficients are at least

$$\frac{139392070260030830922829855080988292511}{111022302462515654042363166809082031250000} > \frac{1}{100}.$$

As in Lemma 5.2, positivity follows from the partition of unity property of the Bernstein basis. The coefficients are computed exactly in the ancillary verifier. $\square$

Equations (10) and (25), together with Lemma 6.1, are contradictory. Thus no counterexample exists for $9/10 \le a \le 1$.

7 Completion of the proof

The ranges $0 \le a \le 9/20$, $9/20 \le a \le 9/10$, and $9/10 \le a \le 1$ have all been excluded under the assumption that every critical point is farther than 1 from a. This proves Theorem 1.1.

A Exact Bernstein verification

For reference, if

$$R(U,V) = \sum_{k=0}^{N} \sum_{\ell=0}^{M} a_{k\ell} U^k V^\ell,$$

then its tensor-product Bernstein coefficients of bidegree (N, M) are

$$b_{ij} = \sum_{k \le i} \sum_{\ell \le j} a_{k\ell} \frac{\binom{i}{k}}{\binom{N}{k}} \frac{\binom{j}{\ell}}{\binom{M}{\ell}}.$$

This identity gives an especially short exact certificate: once all b_{ij} are positive, R is positive on $[0,1]^2$. The accompanying verifier builds the integrals in (14) and (23) symbolically, performs the affine substitutions, applies the displayed conversion formula using rational numbers, and checks the stated coefficient minima. A second implementation independently reconstructs each polynomial from its Bernstein expansion.

References

[1] P. Brändén and D. G. Wagner, A converse to the Grace–Walsh–Szegő theorem, *Math. Proc. Cambridge Philos. Soc.* **147** (2009), 447–453.